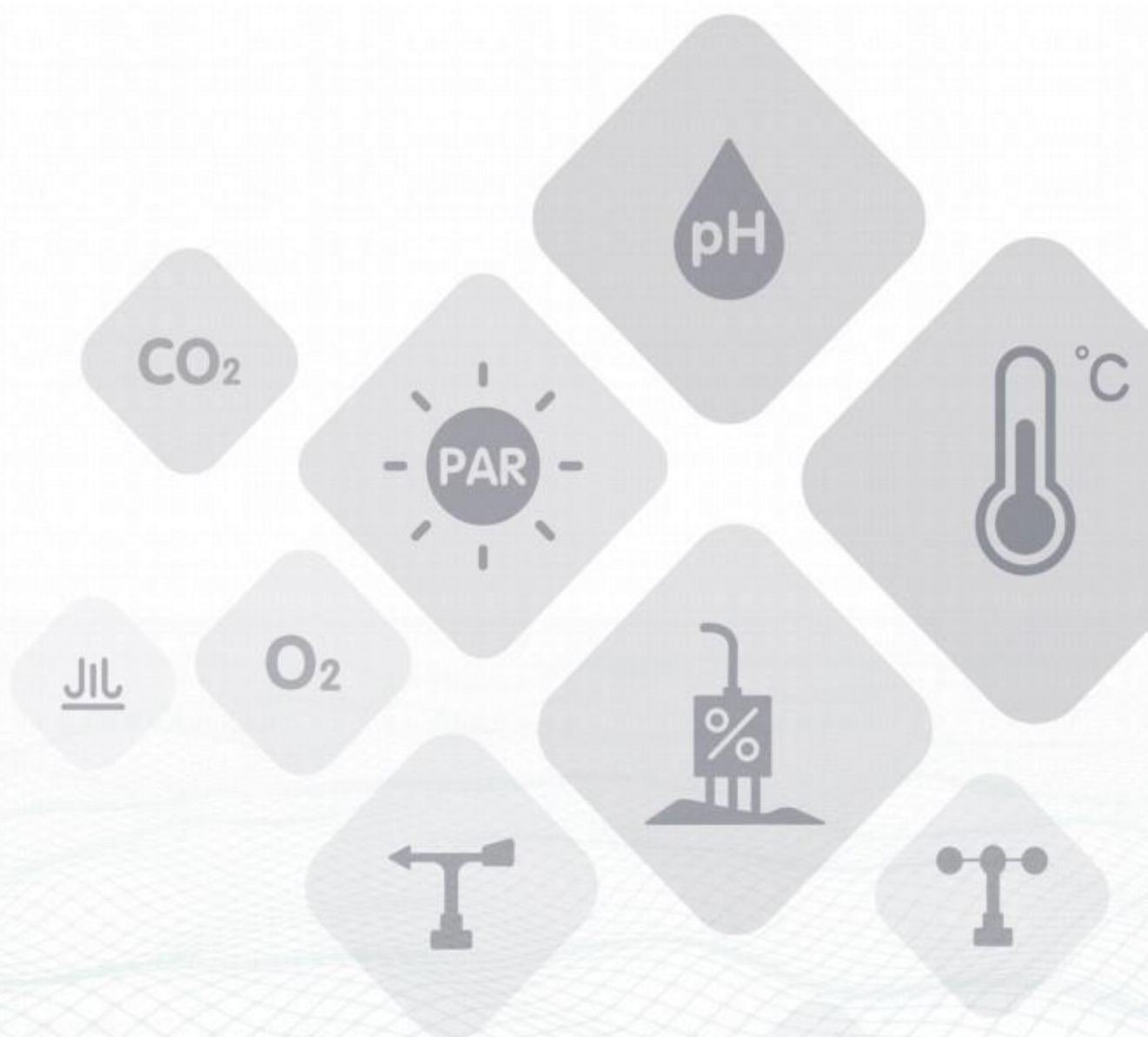




SENSECAP

RS485 Soil & Substrate Moisture, EC, Temperature Sensor- Datasheet

Version: V 1.0



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1. Product Introduction

The Substrate & Soil Moisture, Temperature, and Electrical Conductivity (EC) sensor is an accurate tool for monitoring volumetric water content (VWC), temperature in soil and soilless substrates, and electrical conductivity. It is sealed with resin packaged plastic body with sensing rods which can be inserted directly into the soil or substrate with long time stability.

The sensor is applicable for science research, irrigation, greenhouse, smart agriculture etc.

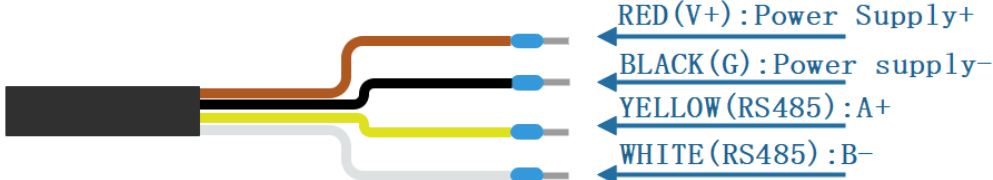
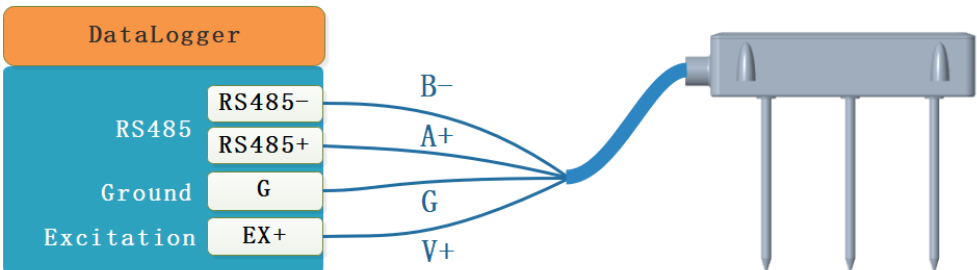
Features

- Integrated with Moisture, EC and Temperature measurement
- Supports Modbus-RTU RS485 communications protocols
- The power input voltage range is wide
- Low-power design supports battery-operated data loggers
- Low salinity sensitivity
- Robust epoxy encapsulation resists corrosive environments
- High accuracy with excellent stability
- Reverse power protection and Built-in TVS/ESD protection

Applications

- Smart Agriculture, Hydroponic and Horticulture
- Greenhouse monitoring
- Soil/Substrate water balance
- Irrigation management
- Solute/fertilizer movement

2. Wiring

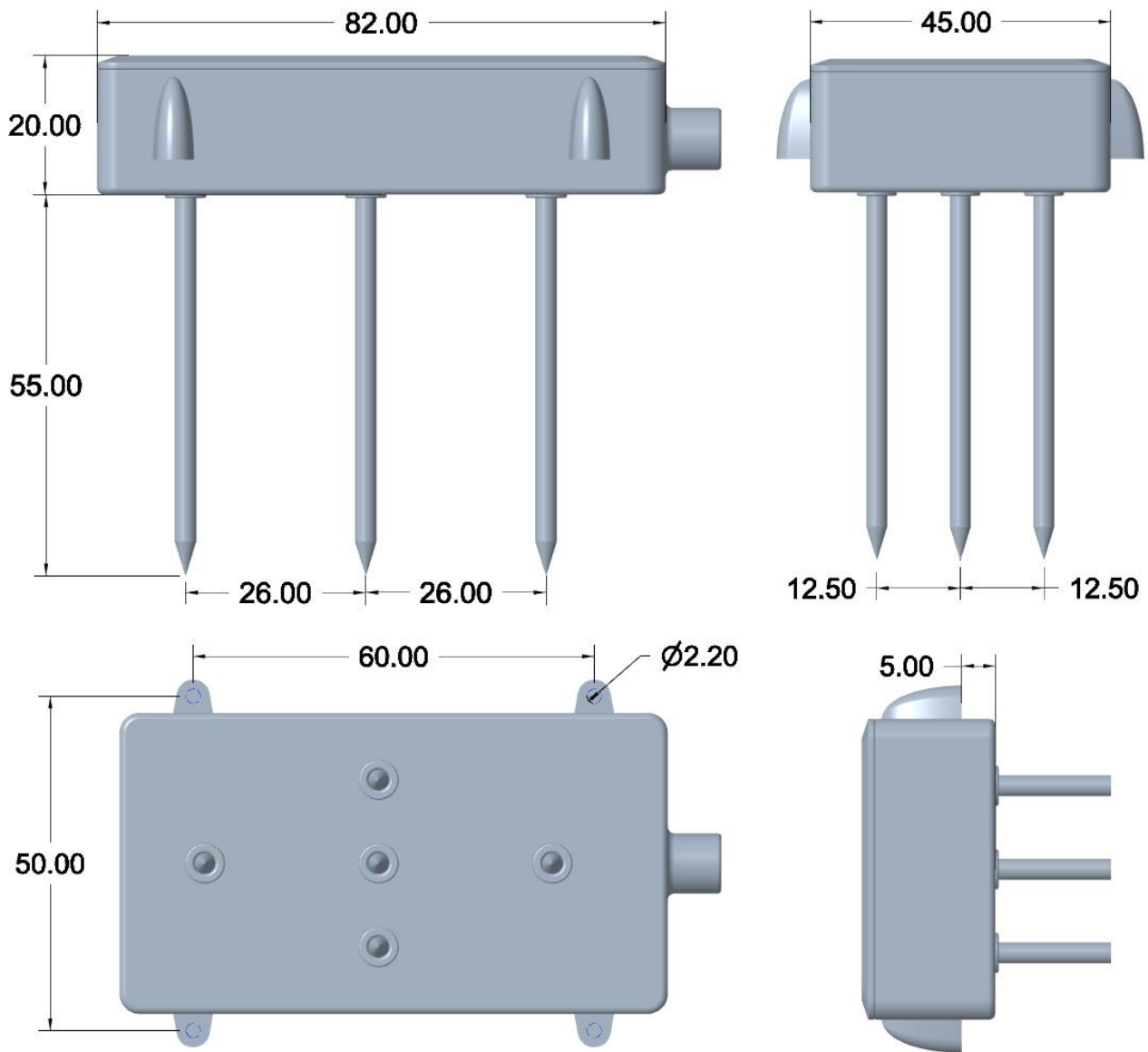
Type	Wiring diagram
RS485 Interface	Cold pressed terminal  RED (V+) : Power Supply+ BLACK (G) : Power supply- YELLOW (RS485) : A+ WHITE (RS485) : B-
Connections	Wiring Diagram  DataLogger RS485- RS485+ Ground Excitation RS485- RS485+ G EX+ B- A+ G V+

3. Specifications

Specifications	
Output Interface	RS485, Modbus-RTU
Power Supply	3.9-28V/DC
Power Consumption	RS485: Quiescent Current: <4mA Measuring Current: <20mA during 150ms measurement
Substrate Moisture Measurement	Apparent dielectric permittivity: Range:1-81 (air - water), Resolution: 1-40.00: 0.1, 40.00-81.88: 0.5 Accuracy:1.00-40.00: +/-1, 40.00-81.00: +/-10% of Readings Volumetric Water Content (VWC): Range:0-100%, Resolution: 0.1% Accuracy: +/-3% The VWC range is dependent on the media and the sensor is calibrated to a custom calibration will accommodate the necessary ranges for most substrates.
Substrate EC Measurement	Range: 0-20ds/m; Resolution: 0.01ds/m; Accuracy: 0-7ds/m, +/-5%; 7-20ds/m, +/-10% of Readings Temperature compensation: 0-50°C
Substrate Temperature Measurement	Range: -40-80°C, Resolution:0.1°C, Accuracy: +/-0.5°C
IP Ratings	IP68
Operating Temperature	-40-80°C
Sensor Sealed	Epoxy resin
Installation	Surface or buried installation
Cable Length	5 meters
Dimension	90*55*75mm

Electrical and Timing Characteristics			
Item	Minimum	Typical	Maximum
Supply Voltage (V+ to G), 3.9-28V DC	3.9 V DC	N/A	28 V DC
Power Line Slew Rate	1.0 V/ms	N/A	N/A
Current Drain (during measurement) , RS485	3.6 mA	4 mA	24mA
Current Drain (while asleep) , RS485	N/A	3.6 mA	N/A
Power-Up Time (RS485)	N/A	300 ms	N/A
Measurement Duration	N/A	150 ms	N/A

4. Dimension



Unit: mm

5. Notes and Installation

5.1 Notes

- The rods of the Sensor are sharp for ease insertion. Care must be taken and precautions followed.
- Avoid touching the rods or exposing them to other sources of static damage, particularly when powered up.
- Do not pull the sensor out of the substrate by its cable.
- If you feel any resistance when inserting the sensor into substrate, Stop pushing and re-insert at a new location.

5.2 Installation

- Clear away any stones. Pre-form holes in very hard soil or substrate before insertion.
- Push the sensor into the substrate until the rods are fully inserted. Ensure good substrate contact.
- If you feel strong resistance when inserting the sensor, you have probably hit a stone. Stop, and re-insert at a new location.

6. RS485 Communication

6.1 Modbus Protocol

Modbus Protocol is widely used to establish master-slave communication between intelligent devices or sensors. A MODBUS message sent from a master to a slave contains the address of the slave, the function code (e.g. 'read register' or 'write register'), the data, and a check sum (LRC or CRC).

The default serial communication settings is slave address 1, Modbus RTU, 9600bps, 8 data bits and 1 stop bit. Following Modbus function code is supported by sensor.

Modbus Function Code 0x03 : used for reading holding register.

Modbus Function Code 0x04 : used for reading input register.

Modbus Function Code 0x06 : used for writing single holding register.

Modbus Function Code 0x10 : used for writing multiple holding register.

6.2 Modbus Register

Parameters	Register Addr. (HEX/DEC)	Data Type	Modbus Function Code(DEC)	Range and Comments	Default Value
SUBSTRATE_TEMPERATURE	0x0000 /0	INT16 RO	3/4	-4000-8000 for -40.00-80.00 (°C); -4000-17600 for -40.00-176.00 (°F)	N/A
SUBSTRATE_VWC	0x0001 /1	INT16 RO	3/4	0-10000 for 0.00-100.00 (%)	N/A
SUBSTRATE_BULK EC	0x0002 /2	INT16 RO	3/4	0-20000 for 0-20000 (us/cm)	N/A
SUBSTRATE_CALIBRATEDRAWCOUNTS	0x0003 /3	UINT16 RO	3/4	0-40960 for 0.0-4096.0	N/A
SUBSTRATE_EPSILON	0x0004 /4	INT16 RO	3/4	0-8188 for 0.00-81.88	N/A
SUBSTRATE_POR EC	0x0005 /5	INT16 RO	3/4	0-20000 for 0-20000 (us/cm)	N/A
SUBSTRATE_TYPE	0x0020 /32	UINT16 RW	3/6/16	0: Soil Media 1: Soilless Media 2: User Defined Curve	0
TEMPERATURE_UNIT	0x0021 /33	UINT16 RW	3/6/16	0: °C 1: °F	0
MEASURE_METHOD	0x0022 /34	UINT16 RW	3/6/16	0: Measure Continuously 1: Measure when	0

				register read command received	
RESERVED	0x0023 /35	UINT16 RW	3/6/16	RESERVED	0
RESERVED	0x0024 /36	UINT16 RW	3/6/16	RESERVED	0
RESERVED	0x0025 /37	UINT16 RW	3/6/16	RESERVED	0
SLAVEADDRESS	0x0200 /512	UINT16 RW	3/6/16	0-255	1
BAUDRATE	0x0201 /513	UINT16 RW	3/6/16	0-5 0:1200bps 1:2400bps 2:4800bps 3:9600bps 4:19200bps 5:38400bps	3:9600bps
PROTOCOL	0x0202 /514	UINT16 RW	3/6/16	0 0:Modbus RTU	0:Modbus RTU
PARITY	0x0203 /515	UINT16 RW	3/6/16	0-2 0:NONE 1:EVEN 2:ODD	0:NONE
DATABITS	0x0204 /516	UINT16 RW	3/6/16	1 1:8 data bits	1:8 data bits
STOPBITS	0x0205 /517	UINT16 RW	3/6/16	0-1 0:1 stop bit 1:2 stop bits	0:1 stop bit
RESERVED	0x0206 /518	UINT16 RW	3/6/16	Reserved	0
RESERVED	0x0207 /519	UINT16 RW	3/6/16	Reserved	0
USERSN	0x0220 /544 0x0221 /545 0x0222 /546 0x0223 /547	UINT16 RW	3/16	0x0000000000000000 000- 0xFFFFFFFFFFFFFFF FF	N/A
SUBSTRATE_TEMPERATURE_FLOAT	0x1000 /4096	FLOAT RO	3/4	-40.00-80.00 (°C); -40.00-176.00 (°F)	N/A
SUBSTRATE_VWC_FLOAT	0x1002 /4098	FLOAT RO	3/4	0.00-100.00 (%)	N/A
SUBSTRATE_BULK_EC_FLOAT	0x1004 /4100	FLOAT RO	3/4	0-20000 (us/cm)	N/A
SUBSTRATE_CALIBRATEDRAWCOUNTS_FLOAT	0x1006 /4102	FLOAT RO	3/4	0.00-4096.00	N/A
SUBSTRATE_EPSILON_FLOAT	0x1008 /4104	FLOAT RO	3/4	0.00-81.88	N/A
SUBSTRATE_PORE_EC_FLOAT	0x100A /4106	FLOAT RO	3/4	0-20000 (us/cm)	N/A
SUBSTRATE_TEMPERATURE_FLOAT	0x1100 /4352	FLOAT _INVE	3/4	-40.00-80.00 (°C); -40.00-176.00 (°F)	N/A

_INVERSE		RSE RO			
SUBSTRATE_VWC _FLOAT_INVERSE	0x1102 /4354	FLOAT _INVE RSE RO	3/4	0.00-100.00 (%)	N/A
SUBSTRATE_BULK EC_FLOAT_INVERSE	0x1104 /4356	FLOAT _INVE RSE RO	3/4	0-20000 (us/cm)	N/A
SUBSTRATE_CALIBRATED DRAWCOUNTS_FLOAT_INVERSE	0x1106 /4358	FLOAT _INVE RSE RO	3/4	0.00-4096.00	N/A
SUBSTRATE_EPSILON FLOAT_INVERSE	0x1108 /4360	FLOAT _INVE RSE RO	3/4	0.00-81.88	N/A
SUBSTRATE_PORE EC_FLOAT_INVERSE	0x110A /4362	FLOAT _INVE RSE RO	3/4	0-20000 (us/cm)	N/A

UINT16:16 bit unsigned integer

INT16:16bit signed integer

RO: Register is Read Only

R/W: Register is Read/Write

HEX: Hexadecimal (data with 0x/0X prefix)

DEC: Decimal

FLOAT: If IEEE754 little-endian byte order is [A,B,C,D], the FLOAT format is [B,A,D,C]

FLOAT_INVERSE: If IEEE754 little-endian byte order is [A,B,C,D], the FLOAT_INVERSE format is [D,C,B,A]

The register value will be set as following when there is sensor error:

Error Value	Register Address	Error
-32768	0x0000-0x000F	Sensor Broken
-32768	0x1000-0x101E	Sensor Broken
-32768	0x1100-0x111E	Sensor Broken
-32765	0x0000-0x000F	Not Supported Value
-32765	0x1000-0x101E	Not Supported Value
-32765	0x1100-0x111E	Not Supported Value

6.3 Modbus Register Detail Description

SUBSTRATE_TEMPERATURE --- Substrate Temperature, INT16 Format		
Range	-4000-8000 for -40.00-80.00 (Temperature Unit set to °C); -4000-17600 for -40.00-176.00 (Temperature Unit set to °F)	Default: N/A

Parameter Storage	N/A	
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SUBSTRATE_TEMPERATURE_FLOAT --- Substrate Temperature, FLOAT Format
SUBSTRATE_TEMPERATURE_FLOAT_INVERSE --- Substrate Temperature, FLOAT_INVERSE Format

Range	-40.00-80.00 (Temperature Unit set to °C); -40.00-176.00 (Temperature Unit set to °F)	Default: N/A
Parameter Storage	N/A	

SUBSTRATE_VWC --- Substrate Volumetric Water Content, INT16 Format

Range	0-10000 for 0-100.00 (%);	Default: N/A
Parameter Storage	N/A	

SUBSTRATE_VWC_FLOAT --- Substrate Volumetric Water Content, FLOAT Format
SUBSTRATE_VWC_FLOAT_INVERSE --- Substrate Volumetric Water Content, FLOAT_INVERSE Format

Range	0-100.00 (%)	Default: N/A
Parameter Storage	N/A	

SUBSTRATE_BULKEC --- Substrate Bulk EC, INT16 Format

Range	0-20000 for 0-20000 (us/cm);	Default: N/A
Parameter Storage	N/A	

SUBSTRATE_BULKEC_FLOAT --- Substrate Bulk EC, FLOAT Format
SUBSTRATE_BULKEC_FLOAT_INVERSE --- Substrate Bulk EC, FLOAT_INVERSE Format

Range	0-20000 (us/cm)	Default: N/A
Parameter Storage	N/A	

SUBSTRATE_CALIBRATEDRAWCOUNTS --- Substrate Water Content Calibrated Raw Counts, UINT16 Format

Range	0-40960 for 0-4096.0;	Default: N/A
Parameter Storage	N/A	

SUBSTRATE_CALIBRATEDRAWCOUNTS_FLOAT --- Substrate Water Content Calibrated Raw Counts, FLOAT Format
SUBSTRATE_CALIBRATEDRAWCOUNTS_FLOAT_INVERSE --- Substrate Water Content Calibrated Raw Counts, FLOAT_INVERSE Format

Range	0-4096.00	Default: N/A
Parameter Storage	N/A	

SUBSTRATE_EPSILON --- Substrate Epsilon, UINT16 Format		
Range	0-20000 for 0.00-200.00	Default: N/A
Parameter Storage	N/A	

SUBSTRATE_EPSILON_FLOAT --- Substrate Epsilon, FLOAT Format		
SUBSTRATE_EPSILON_FLOAT_INVERSE --- Substrate Epsilon, FLOAT_INVERSE Format		
Range	0.00-200.00	Default: N/A
Parameter Storage	N/A	

SUBSTRATE_POREEC --- Substrate Pore EC, INT16 Format		
Range	0-20000 for 0-20000 (us/cm);	Default: N/A
Parameter Storage	N/A	

SUBSTRATE_POREEC_FLOAT --- Substrate Pore EC, FLOAT Format		
SUBSTRATE_POREEC_FLOAT_INVERSE --- Substrate Pore EC, FLOAT_INVERSE Format		
Range	0-20000 (us/cm)	Default: N/A
Parameter Storage	N/A	

SUBSTRATE_TYPE--- Substrate Type, INT16 Format		
Data Range	0: Soil Media 1: Soilless Media 2: User Defined Curve	Default: 1
Parameter Storage	YES	

TEMPERATURE_UNIT--- Temperature Unit, INT16 Format		
Data Range	0:°C 1:°F	Default: 0
Parameter Storage	YES	

MEASURE_METHOD--- Measure Method, INT16 Format		
Data Range	0: Measure continuously 1: Measure on register reading command received	Default: 1
Parameter Storage	YES	

SLAVEADDRESS --- Modbus Slave Address (16 bit unsigned int)		
Data Range	0-255	Default: 1
Parameter Storage	YES	

Note: Please power up the sensor to take effective after set.

BAUDRATE --- Serial Comm Baudrate (16 bit unsigned int)		
Data Range	0-5 0:1200bps 1:2400bps 2:4800bps 3:9600bps 4:19200bps 5:38400bps	Default: 3
Parameter Storage	YES	

Note: Please power up the sensor to take effective after set.

PROTOCOL --- Serial Comm Protocol (16 bit unsigned int)		
Data Range	0 0:Modbus RTU	Default: 0
Parameter Storage	YES	

Note: Please power up the sensor to take effective after set.

PARITY --- Serial Comm Parity (16 bit unsigned int)		
Data Range	0-2 0:NONE 1:EVEN 2:ODD	Default: 0
Parameter Storage	YES	

Note: Please power up the sensor to take effective after set.

DATABITS --- Serial Comm Databits (16 bit unsigned int)		
Data Range	1 1:8 databits	Default: 1
Parameter Storage	YES	

Note: Please power up the sensor to take effective after set.

STOPBITS --- Serial Comm Stop bits (16 bit unsigned int)		
Data Range	0-1 0:1 stop bit 1:2 stop bits	Default: 0
Parameter Storage	YES	

Note: Please power up the sensor to take effective after set.

USER_SN--- User Serial Number		
Data Range	0x0000000000000000-0xFFFFFFFFFFFFFFFF User Serial Number. These four registers should be read and write together from start address 0x0220 with quantity 4.	Default: N/A
Parameter Storage	YES	

6.4 Modbus Function Code

For description below, data started with 0X/0x means that it's in HEX format.

6.4.1 Function Code 3 Protocol Example

Master Request: AA 03 RRRR NNNN CCCC

AA	1 byte	Slave Address,0-255
0x03	1 byte	Function Code 3
RRRR	2 byte	Starting Register Addr.
NNNN	2 byte	Quantity of Register to read
CCCC	2 byte	CRC CHECKSUM

Slave Response: AA 03 MM VV0 VV1 VV2 VV3... CCCC

AA	1 byte	Slave Address,0-255
0x03	1 byte	Function Code 3
MM	1 byte	Register Data Byte Count
VV0,VV1	2 byte	Register Value (High8bits first)
VV2,VV3	2 byte	Register Value (High8bits first)
...	...	Register Value (High8bits first)
CCCC	2 byte	CRC CHECKSUM

Example: Read register 0x0200-0x0201,that is slave address and baud rate.

Master Request:01 03 0200 0002 C5B3

Slave Addr.	1 byte	0x01
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Function Code	1 byte	0x03
Starting Register Addr.	2 byte	0x0200
Quantity of Register to read	2 byte	0x0002
Checksum	2 byte	0xC5B3

Slave Response: 01 03 04 00 01 00 03 EB F2

Slave Addr.	1 byte	0x01
Function Code	1 byte	0x03
Register Data Byte Count	1 byte	0x04
Register Value: Address	2 byte	0x00(HIGH 8 Bits)
		0x01(LOW8 Bits)
Register Value: Baud rate	2 byte	0x00(HIGH 8 Bits)
		0x03(LOW8 Bits)
Checksum	2 byte	0xEBF2

6.4.2 Function Code 4 Protocol Example

Master Request: AA 04 RRRR NNNN CCCC

AA	1 byte	Slave Address,0-255
0x04	1 byte	Function Code 4
RRRR	2 byte	Starting Register Addr
NNNN	2 byte	Quantity of Register to read
CCCC	2 byte	CRC CHECKSUM

Slave Response: AA 04 MM VV0 VV1 VV2 VV3... CCCC

AA	1 byte	Slave Address,0-255
0x04	1 byte	Function Code 4
MM	1 byte	Register Data Byte Count
VV0,VV1	2 byte	Register Value (High8bits first)
VV2,VV3	2 byte	Register Value (High8bits first)
...	...	Register Value (High8bits first)
CCCC	2 byte	CRC CHECKSUM

Example: Read register 0x0000-0x0003, that is substrate temperature, VWC, EC, calibrated Raw Counts.

Master Request: 01 04 0000 0004 F1C9

Slave Addr.	1 byte	0x01
Function Code	1 byte	0x04

Starting Register Addr.	2 byte	0x0000
Quantity of Register to read	2 byte	0x0003
Checksum	2 byte	0xF1C9

Slave Response: 01 04 08 08 02 12 E8 05 32 70 B5 20 7A

Slave Addr.	1 byte	0x01
Function Code	1 byte	0x04
Register Data Byte Count	1 byte	0x08
Register Value: Substrate Temperature	2 byte	0x08(HIGH 8 Bits)
		0x02(LOW8 Bits)
Register Value: Substrate VWC	2 byte	0x12(HIGH 8 Bits)
		0xE8(LOW8 Bits)
Register Value: Substrate EC Bulk	2 byte	0x05(HIGH 8 Bits)
		0x32(LOW8 Bits)
Register Value: Substrate Calibrated Raw Counts	2 byte	0x70(HIGH 8 Bits)
		0xB5(LOW8 Bits)
Checksum	2 byte	0x207A

6.4.3 Function Code 6 Protocol Example

Master Request: AA 06 RRRR VVV CCCC

AA	1 byte	Slave Address,0-255
0x06	1 byte	Function Code 6
RRRR	2 byte	Register Addr (High8bits first)
VVV	2 byte	Register Value (High8bits first)
CCCC	2 byte	CRC CHECKSUM

Slave Response: AA 06 RRRR VVV CCCC

AA	1 byte	Slave Address,0-255
0x06	1 byte	Function Code 6
RRRR	2 byte	Register Addr (High8bits first)
VVV	2 byte	Register Value (High8bits first)
CCCC	2 byte	CRC CHECKSUM

Example: Write Register 0x0200,that is change Modbus slave address to 2.

Master Request: 01 06 0200 0002 09B3

Slave Addr.	1 byte	0x01
Function Code	1 byte	0x06
Register Addr.	2 byte	0x0200

Register Value	2 byte	0x0002
Checksum	2 byte	0x09B3

Slave Response: 01 06 0200 0002 09B3

Slave Addr.	1 byte	0x01
Function Code	1 byte	0x06
Register Addr.	2 byte	0x0200
Register Value	2 byte	0x0002
Checksum	2 byte	0x09B3

6.4.4 Function Code 16 Protocol Example

Master Request: AA 10 RRRR NNNN MM VVV1 VVV2 ...CCCC

AA	1 byte	Slave Address,0-255
0x10	1 byte	Function Code 0x10
RRRR	2 byte	Starting Register Addr
NNNN	2 byte	Quantity of Register to write
MM	1 byte	Register Data Byte Count
VVV1	2 byte	Register Value(High8bits first)
VVV2	2 byte	Register Value(High8bits first)
...	...	Register Value(High8bits first)
CCCC	2 byte	CRC CHECKSUM

Slave Response: AA 10 RRRR NNNN CCCC

AA	1 byte	Slave Address,0-255
0x10	1 byte	Function Code 0x10
RRRR	2 byte	Starting Register Addr
NNNN	2 byte	Quantity of Register to write
CCCC	2 byte	CRC CHECKSUM

Example: Write Register 0x0200-0x0201, that is set slave address to 1, and baud rate to 19200bp.

Master Request:01 10 0200 0002 04 0001 0004 BACC

0x01	1 byte	Slave Addr.
0x10(HEX)	1 byte	Function Code 0x10
0x0200	2 byte	Starting Register Addr
0x0002	2 byte	Quantity of Register to write
0x04	1 byte	Register Data Byte Count
0x0001	2 byte	Register Value: Slave Address 1
0x0004	2 byte	Register Value: Baud rate 19200bps

0xBACC	2 byte	CRC CHECKSUM
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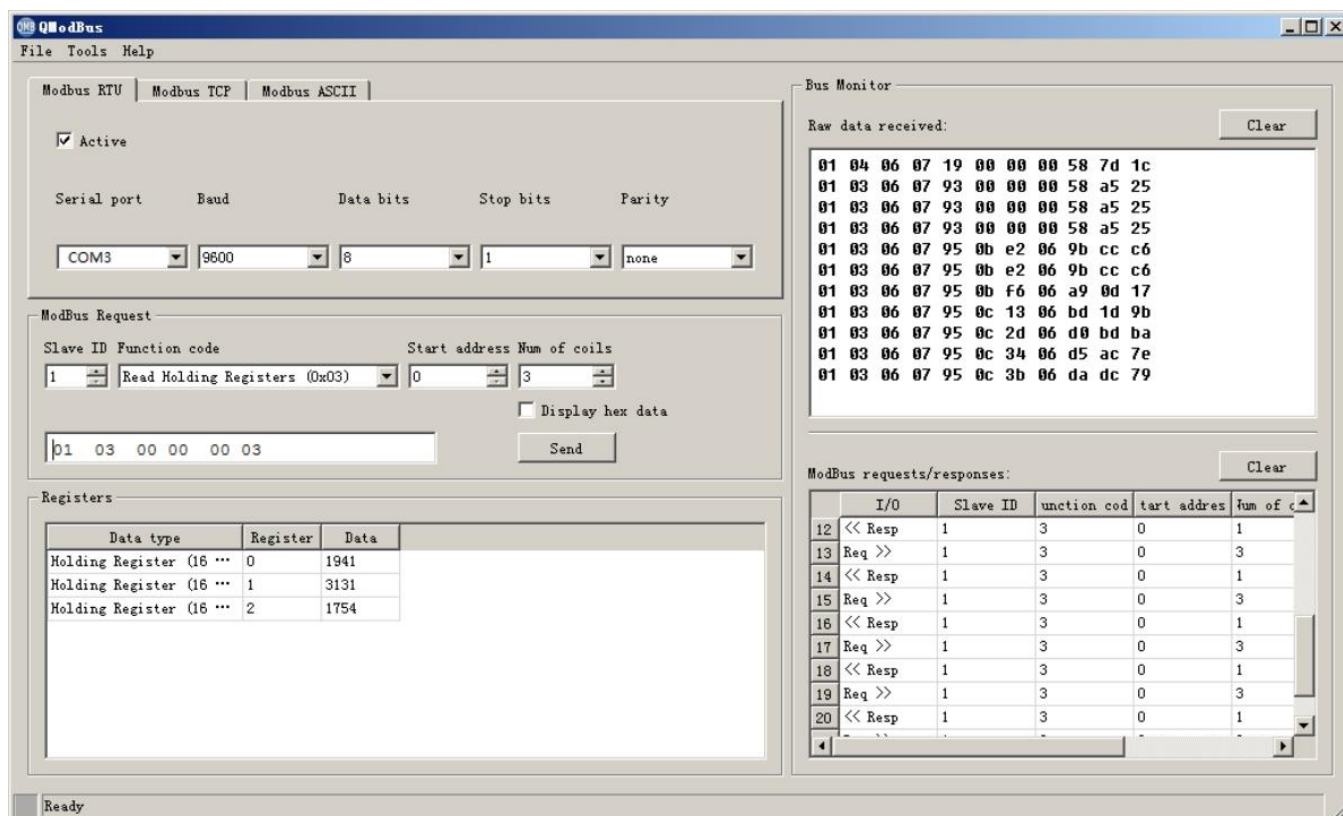
Salve Response: 01 10 0200 0002 4070

0x01	1 byte	Slave Addr.
0x10(HEX)	1 byte	Function Code 0x10
0x0200	2 byte	Starting Register Addr(High8bits first)
0x0002	2 byte	Quantity of Register to write(High8bits first)
0x4070	2 byte	CRC CHECKSUM

6.5 Software Configuration Utility

6.5.1 Universal Modbus Comm Utility

You can use software listed below to try reading/writing the register of sensor,
<https://github.com/ed-chemnitz/qmodbus/releases>



7. Version

V1.0	2024-08-12	First release	Yvonne
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